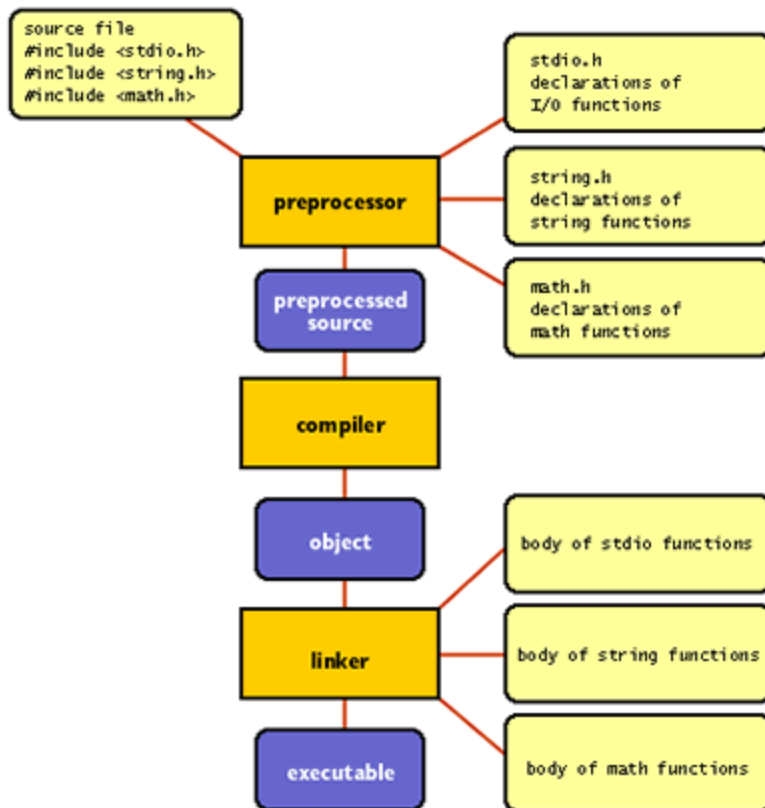


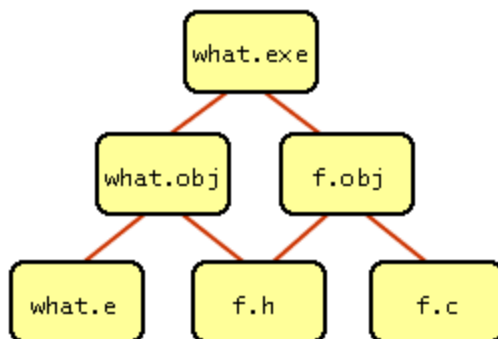
Topic 2.6

2.6: C Header File and Lib File Organization



Dependency diagrams and the make utility

In the *what.c* program, there are a number of dependencies, illustrated by the following dependency graph:



If something in this graph changes, all the items *above* it need to be recompiled and relinked.

In UNIX, the *make* utility simplifies the task of recreating an executable when something changes. The format of a command in *make* is

```
<tab>target: <tab><tab>dependent-list
<tab><tab><tab>command
```

where

Make Utility

<i>target</i>	specifies an object module, executable, etc.
<i>dependent-list</i>	is a list of objects on which the target depends
<i>command</i>	specifies how to recreate the target

There must be a tab after *target:* and before *command*.

The commands are placed in a file with default name *makefile*.

To invoke *make*:

Make

make		# Uses default file <i>makefile</i> .
make	-f mf_name#	Use named makefile <i>mf_name</i> .

There are many default rules for how to make objects from Fortran sources, C sources, etc. Also, the *project* feature of most integrated environments is based on this idea of a make file.

Makefile for what.c

```
# This is a makefile for Borland C on a PC using DOS.

what.exe:<tab>what.obj <tab>f.obj
```

```
<tab><tab>tcc <tab><tab>-ewhat.exe <tab>what.obj <tab>f.obj  
  
what.obj:<tab>what.c <tab><tab>f.h  
<tab><tab>tcc <tab><tab>-c <tab>what.c  
  
f.obj:<tab><tab>f.c <tab>f.h  
<tab><tab>tcc <tab>-c <tab>f.c
```

Topic 2.7 (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-2-dot-7>)